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Coupled Lab-Scale and Numerical Analysis of Proppant Transport in Unconventional Reservoir

Lei Mu and Jie Yang, Luliang Oilfield Operation Area, Xinjiang Oilfield Branch, PetroChina, Karamay, China; Changwei Wang, Development Division, Xinjiang Oilfield Branch, PetroChina, Karamay, China; Xingning Huang and Xi Yan, CK Energy Technology Co., Ltd, Chengdu, China

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Abstract

Constant-proppant-concentration fracturing offers an efficient solution for China's unconventional resources, addressing excessive fluid consumption and inconsistent placement in conventional stepwise designs. Validation via geo-engineering integrated software and fractal-based rough fracture modeling demonstrates superior proppant transport for 20/40, 40/70, and 70/140 mesh sands: conventional methods form near-wellbore dunes with sparse distal coverage, while constant concentration achieves uniform distribution along fracture length, enhancing toe-end concentration, effective fracture support, and hydrocarbon mobility. This technique optimizes placement profiles and stimulation economics. Proppant distribution within fractures is a crucial factor in enhancing the production from fracturing. Currently, most studies on the transport and settling behavior of proppants in fractures are based on the models of smooth, straight fracture. There is rarely research considering the effects of fracture tortuosity and roughness on proppant movement and settling. To further reveal the trends of proppant transport and setting in tortuous and rough fractures, physical experiments were conducted to observe proppant behavior in smooth and flat fractures firstly. Based on this, established tortuous and rough fracture models to explore proppant movement and settling patterns. Results indicate that larger proppant particle sizes are more conducive to settling within smooth plate fractures, and medium-to-large particle proppants are more likely to accumulate near the branch fracture entrances. For smooth-plate fractures, the increased the injection rates will allow proppants to penetration deeper into both main and branch fractures. For tortuous rough fractures, larger proppants are more impacted by fracture structure, often accumulating in the middle sections and causing blockage; however, increased injection speeds can effectively mitigate these blockages. These findings reveal the influence of various operational factors on proppant transport and settling, offering theoretical guidance for optimizing hydraulic fracturing techniques.

Introduction

Hydraulic fracturing is a key technology for efficiently developing unconventional resources such as shale gas, shale oil, and tight oil/gas reserves. Understanding the distribution patterns of proppants within

hydraulic fractures is fundamental for optimizing sand-laden fracturing, creating high-conductivity flow channels. To investigate proppant distribution and settling within fractures, extensive research has been conducted globally through laboratory physical models and numerical simulations. Most models were based on the simplified KGD fracture model. The fracture was simplified as a rectangular prism, omitting variations in fracture width and height along the length. Early studies used parallel glass plates to model fractures and studied proppant distribution and settling behavior, observing the formation of sand beds as proppants settled after entering fractures, with subsequent proppants moving deeper into the fracture driven by the fluid. Researchers such as Malhotra and Wen Qingzhi utilized large, visual plate setups to simulate artificial fractures to exam how factors like fluid viscosity, fracture width, and sand-to-fluid ratios impact proppant distribution and settling, and established sensitivity analyses for these variables. Besides these factors, they also studied the impact of perforation parameters and fracture morphology. During volume fracturing, fractures often form as networks, and Wen Qingzhi and others used complex "T"- and "H"-type models to study proppant movement within these networks, proposing that dune formation in branch fractures is the result of both sand carrying force and gravity. Noting that rough fracture walls could hinder proppant movement, some scholars employed 3D printing to simulate rough fracture surfaces, increasing wall roughness in both single and cross fractures. Findings revealed more even proppant distribution within rough fractures, whereas proppants in smooth fractures tended to settle unevenly under turbulent flow conditions. With the development of hydraulic fracturing, it is increasingly recognized that fracture exhibit a tortuous distribution pattern during their propagation. However, constructing physical models of tortuous fractures remains complex, and most studies on proppant transport within such fractures rely on numerical simulations. Common numerical methods include Euler-Lagrange and Euler-Euler approaches. The Euler-Lagrange method treats fluids as a continuous phase, calculated using the Euler method, and proppant particles as a discrete phase, calculated via the Lagrange method, typically coupled through CFD-DEM. This method, however, is resource-intensive and time-consuming, making it unsuitable for large-scale fracture models. In contrast, the Euler-Euler method treats solid particles as a pseudo-fluid, with mutual penetration between liquid and solid phases, and volume relationships between phases that sum to 1, allowing for efficient analysis of proppant transport and dune formation. Using numerical simulations can circumvent the complex construction of physical models, facilitating the modeling of diverse, complex fracture networks. Numerical simulation is becoming a primary approach to studying proppant behavior in intricate fracture networks. However, most proppant transport studies use single-method approach and rarely consider tortuosity and wall roughness effects currently. This study integrates physical experiments and numerical simulations to explore proppant transport and settling patterns in fractures. By first simulating smooth, straight fracture transport behavior through physical experiments, we establish a numerical simulation model using the Euler-Euler method, validating simulation results against experimental findings. We further investigate the transport and settling patterns in tortuous rough fractures, analyzing the effect of different factors, including dune morphology, supporting area, and height changes, providing theoretical support for volumetric fracturing design. Constant-proppant-concentration fracturing presents a technically viable solution for unconventional reservoirs in China, addressing limitations of conventional stepwise-concentration methods-specifically excessive fluid consumption and inconsistent proppant placement. This approach targets challenges in tight/shale reservoirs, including low pressure, low permeability, and poor fluid recovery. Proppant transport in complex fracture networks is governed by fracture geometry, proppant properties (density, size), fluid characteristics, and placement methods. While conventional stepwise designs (post-Nolte 1986) optimize pad volume via fluid efficiency analysis, they exhibit poor distal proppant distribution. Chevron's Permian Basin implementation and Singh et al.'s research indicate constant-concentration fracturing reduces friction, minimizes gel usage, and mitigates screenout risks, though its impact on conductivity and production remains underexplored for diverse reservoirs. Leveraging Petrel-based geo-engineering modeling, this study contrasts proppant transport dynamics between constant-concentration and stepwise methods, evaluating particle size/density effects

on placement efficiency. Findings guide optimal proppant selection and design strategies to achieve full-fracture coverage and enhanced stimulation effectiveness.

Overview

Fractures, ubiquitous structural features in rock formations, critically impact subsurface stability and engineering outcomes. In petroleum extraction, fracture initiation and propagation compromise reservoir integrity, adversely affecting hydrocarbon productivity. Traditional characterization oversimplifies fractures as geometrically regular surfaces, neglecting intrinsic complexity. Fractal theory reveals their inherent self-similarity and scale invariance, enabling accurate morphological and evolutionary modeling. This paradigm shift facilitates robust rough fracture models that replicate true fracture networks through fractal algorithms, capturing structural heterogeneity and distribution patterns. Such models provide critical insights into fracture mechanics and growth prediction while supporting engineering applications: they quantify proppant transport efficiency in hydraulically fractured wells by simulating fracture connectivity and spatial configuration, optimize stimulation parameters, and derive fracture porosity/permeability to predict reservoir flow behavior and production capacity. Ultimately, fractal-based modeling advances fracture diagnostics and stimulation design for enhanced recovery.

Fractal theory-based rough fracture modeling has gained significant traction in geological engineering research, enabling quantitative characterization of fracture networks through fractal dimensions that evaluate reservoir properties under varying geological conditions. These models critically predict formation permeability and porosity, directly informing hydrocarbon exploration and hydrogeological studies. Crucially, fractal fracture morphology substantially influences proppant behavior: multi-scale interfacial contacts between proppant particles and fracture surfaces govern placement efficiency and consolidation effectiveness. Research confirms that fractal geometry induces non-uniform proppant distribution-particles preferentially accumulate at fracture edges and asperities, compromising placement uniformity. Conversely, increased surface roughness enhances proppant-rock contact area, improving consolidation integrity. Fractal dimensions provide quantitative roughness parameters essential for simulating proppant transport dynamics and optimizing placement strategies.

Geological profile

Formation depositional characteristics

In the study area, the target layer of Triassic Baikouquan formation was angular unconformable deposits above the Permian Lower Urho Formation. The formations from bottom to top of Da13 encountered are the Permian Lower Urho Formation, the Triassic Baikouquan Formation, Karamay Formation, Baijiantan Formation, Jurassic Badaowan Formation, Sangonghe Formation, Xishanyao Formation, Toutunhe Formation, Qigu Formation, and Cretaceous Formation. Notably, the contacts between the Permian and Triassic, Triassic and Jurassic, and Jurassic and Cretaceous formations are regional unconformable contacts.

Based on lithological and electrical characteristics, the Triassic Baikouquan Formation in this region is subdivided from bottom to top into three sections: Bai1 (T_1b_1), Bai2 (T_1b_2), and Bai3(T_1b_3). In the drilling area of Well 13, the Baikouquan Formation exhibits a thickness variation trend towards the center of the Mahu Depression, and the strata gradually thins out in the upward-dipping north-eastern direction. The thickness of the Baikouquan Formation is 119.9~177.6m, T_1b_3 and T_1b_2 are relatively stable, the thickness of T_1b_3 is 55.4m~63.3m with an average of 56.8m, T_1b_2 is 51.7m~67.2m with an average of 59.3m. Due to the superimposed sedimentation at the base of Baikouquan, the thickness of T_1b_1 varies significantly from southwest to northeast (higher on the slope). The thickness of T_1b_1 is 16.7~60.6m with an average of 45.2m(Figure 1).

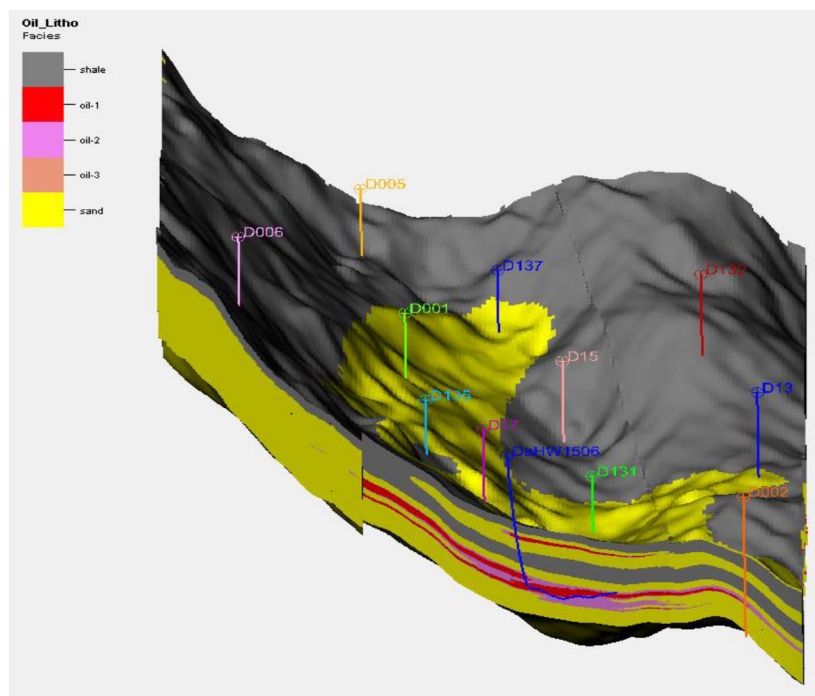


Figure 1—3D modeling for study area

Lithology characteristics

The Triassic Baikouquan Formation is mainly composed of gray medium-grained conglomerate, sandy fine-grained conglomerate, sandy small-grained conglomerate, and gravelly coarse sandstone, followed by conglomerate-bearing siltstone, gray-brown mudstone, and silty mudstone. Specifically, T_1b_1 is dominated by gray and brown medium-grained conglomerate, locally interbedded with brown and gray-brown mudstone, conglomerate-bearing mudstone and muddy siltstone. Electrical characteristics show that the bilaterally oriented resistivity is low. In the lower part of T_1b_2 , is gray massive muddy medium-grained conglomerate which is denser and with medium-low-resistance, and the upper part consists of gray blocky sandy fine-grained conglomerate, sandy small-grained conglomerate, and conglomerate-bearing medium to coarse sandstone, interbedded with brown blocky mudstone, exhibiting medium-high-resistance blocky and finger-like features. T_1b_3 is dominated by brown mudstone with interbedded thin layers of gray muddy siltstone, displaying low-resistance finger-like and local serrated features. From bottom to top in the Baikouquan Formation, the overall rock grain size gradually becomes finer, and mudstone develops and thickens. This variation is clearly reflected in the natural gamma curve.

In Da13 well, the lithology of Baikouquan Formation is mainly gray sandy fine-grained conglomerate and sandy small-grained conglomerate, followed by medium-coarse sandstone and fine sandstone, with fluorescence, oil traces and oil spots. Core samples from T_1b_2 is dominated by sandy fine-grained conglomerate and sandy small-grained conglomerate, accounting for 62.9%, followed by medium-coarse sandstone (8%) and fine sandstone (9.2%). The oil content is closely related to lithology, the oil content is concentrated in fine-grained conglomerate and small-grained conglomerate, followed by fine sandstone and medium-coarse sandstone, while other lithologies are with fewer oil content. Under the microscope, the structure of the core is mainly characterized by a sandy conglomerate structure, primarily supported by particles. The gravel size varies, generally ranging from 2 to 16 mm, with moderate to good sorting. The rock particles are mainly subrounded, occasionally subangular. The gravel composition is complex, primarily consisting of tuff, followed by felsite and andesite. The sandy component is mainly tuff, followed by quartz and feldspar, with small amounts of felsite, granite and andesite. The matrix material has a low content, accounting for 3.5%, mainly composed of clay, followed by kaolinite. The cement content is low,

only 1%, mainly composed of calcite. The cement is dense, and the grain contact is dominated by point-line and line contacts. The cementation include mosaic and pore-mosaic, with low maturity in both composition and structure. Mild to moderate feldspar alteration and secondary enlargement of quartz are commonly observed in the rock.

Storage space of reservoir

Based on observations of thin-section microscopy, the predominant pores in the T_1b_2 are intergranular pores, followed by intra-granular dissolution pores, cast pores, and a minor amount of residual intergranular pores. Fluorescence thin sections reveal that hydrocarbons are primarily present in the primary intergranular pores and intra-granular dissolution pores (Figure 3). According to the statistical analysis of whole-rock samples from the T_1b_2 core from well D13, the total clay mineral content is from 1.07% to 7.62%, with an average of 3.38%. The clay mineral composition is dominated by illite-montmorillonite mixed layer and chlorite (Figure 2).

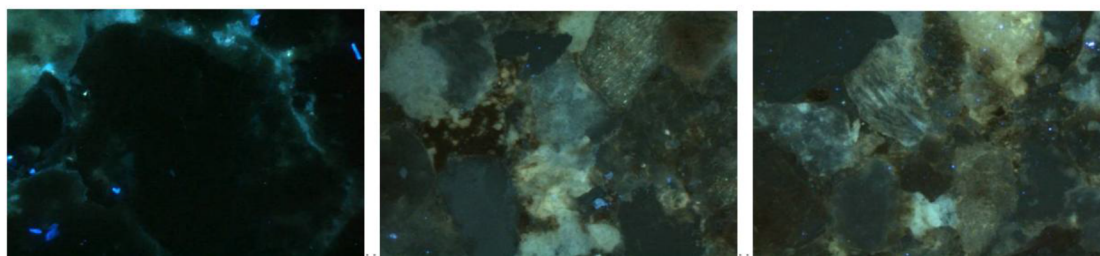


Figure 2—Fluorescence thin section of T_1b_2 core

The capillary pressure curve of the T_1b_2 core in D13 exhibits a skewed fine-skewness profile, indicating that the reservoir has a predominant high displacement pressure with predominantly small to medium pore throats. There are also high-quality reservoirs with low displacement pressure and large pore throat radii. In general, 60% of the samples have a maximum mercury injection volume of less than 50%, and the mercury withdrawal efficiency is relatively low (averaging 25%). This suggests that the overall physical properties of the block are poor, and the pore throats are relatively large.

Comparatively, when compared to other tight conglomerate reservoirs in the Mahu area, the T_1b_2 reservoir in D13 reflects a poorer pore structure in the mercury injection experiment. The displacement pressure and median pressure are relatively high, indicating poor interconnectivity between pores in this reservoir zone.

Natural fracture characteristics

There are 11 coring wells in the Baikouquan Formation in the Da13 area, total core length is 219.6 meters. no natural fractures were observed on the core samples (Figure 3). There are 5 wells with resistivity imaging logging data in this area (Da002, Da005, Mazhong2, Mazhong5, and Yanbei2). Only Da002 and Da005 logging data show a small amount of fractures in the Baikouquan Formation. Based on the coring and image logging interpretation concluded that natural fractures in the Baikouquan Formation are not developed.



Fracturing Parameter Selection and Treatment Design

Fracturing parameters were designed based on Sulige Gas Field tight gas pilot wells: injection rates of 5-8 m³/min per well, proppant volumes of 30-60 m³ per stage, and 20/40-, 40/70-, and 70/140-mesh quartz sand. Conventional stepwise simulations employed six pumping stages with 300 m³ total fluid per stage,

while constant-concentration designs used three stages with 250 m³ fluid per stage. Simulation parameters are detailed in Table 1.

Table 1—Operational Parameter Comparison: Constant-Concentration vs. Stepwise Proppant Placement

method	quartz sand mesh	quartz sand volume m ³	injection rates m ³ /min	proppant volume m ³
stepwise	20/40, 40/70, 70/140	30, 40, 50, 60	5, 6, 7, 8	300
constant	20/40, 40/70, 70/140	30, 40, 50, 60	5, 6, 7, 8	250

Constant proppant concentration here refers to rapidly ramping up concentration during initial injection and sustaining that level until completion, rather than maintaining uniform concentration throughout the entire fracturing operation. This approach intentionally reduces total fluid volume per stage compared to conventional methods to highlight its potential advantages. Operational distinctions between stepwise and constant-concentration designs are illustrated in Fig. 1 under an 8 m³/min injection rate.

Transport and Placement Patterns of Different Proppants Under Two Placement Methods

With fixed geomechanical and operational parameters, proppant placement mechanisms for 20/40-, 40/70-, and 70/140-mesh quartz sand were investigated under four proppant volumes and injection rates per stage. Proppant coverage distribution along primary fractures was characterized, comparing differences between stepwise and constant-concentration methods at equivalent proppant volumes to reveal their distinct transport and placement dynamics.

Proppant Transport and Placement. For 20/40-mesh quartz sand at 60 m³ per stage, stepwise fracturing increased proppant concentration incrementally from 7% to 32%, while constant-concentration fracturing ramped from 12% to 29% over three stages with lower peak concentration. Table 3 displays proppant distribution and fracture width under identical injection rates post-stimulation. Stepwise methods generated longer fractures but formed two continuous proppant dunes near the wellbore with sparse distal coverage. Constant-concentration designs produced balanced dual-wing fractures with uniform proppant distribution from near-well to toe regions, enhancing overall conductivity and proppant support efficiency. This optimized placement profile improves hydrocarbon migration and production potential.

Fracturing design and optimization. In the Da13 well area, the Kexia Formation reservoir is characterized by low porosity, extra-low permeability, and high fluid flow resistance within the formation. To enhance single well production, reservoir modification is essential to expand the effective area and volume of production pressure. Considering the geological characteristics of the reservoir and the current application status of horizontal well fracturing technology in Xinjiang oil fields, the following fracturing design principles are proposed: The main oil reservoirs in Da13 area are with low porosity and extra-low permeability, no natural productivity after initial vertical well perforation, large-scale hydraulic fracturing is necessary to achieve oil production. The reservoir conditions are suitable for large-scale hydraulic fracturing. The main sand layer is located in T₁b₂, with a reservoir thickness of 38.8 to 44.0 meters and oil layer thickness of 18.0 to 29.0 meters. Oil and gas shows are concentrated in T₁b₂, with a reservoir thickness of 21.6 to 25.0 meters and oil layer thickness of 13.0 to 26.0 meters. Considering natural fractures in the reservoir are not developed, the stress difference between the two-phase is large, the brittleness index is medium, low likelihood of forming complex fractures during fracturing, as so cluster perforation per segment aim to increase the modification volume and fully produce the sweet spots of reservoir.

Hydraulic Fracturing program design

The hydraulic fracturing program takes DaHW well as an example, with a horizontal section length of 1800.0m. The drilling objective is to develop the Baikouquan Formation reservoir in the Da13 area of the

Mahu oil field, supplement the production capacity of the well area, and ensure production stability. The actual drilling fluid usage aligns with the design, with a horizontal section drilling fluid density ranging from 1.40 to 1.43 g/cm³. Five segments in the horizontal section are with abnormal gas testing results, one segment indicates gas stained, and the peak value of the gas measurement is 33.07%. The actual drilling trajectory profile is shown as Figure 5.

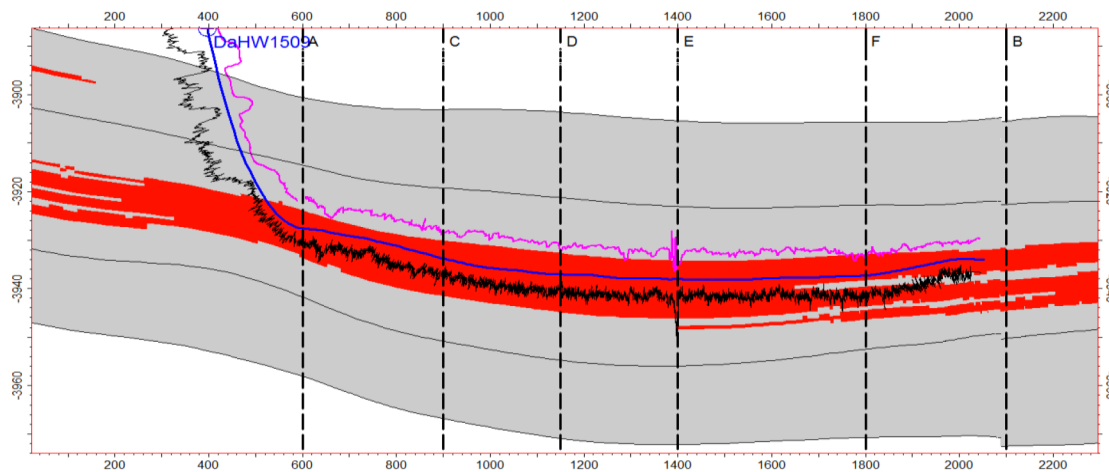


Figure 5—Well actual drilling trajectory vs reservoir profile

Proppant type recommendation. Post-fracturing, the oil layers is cracked, allowing oil and gas to aggregate and flow out from the fracture channels, and the high pressure leading to the deformation around the wellbore and fracture channels forming. These channels possess high flow capacity. In actual production, to prevent fractures from closing and to ensure the smooth passage of petroleum, proppant are introduced into the high-pressure solution. This provides support to the fractures, preventing closure due to stress release. By maintaining a high flow capacity, it facilitates the unimpeded flow of oil and gas, increasing production. This approach, to some extent, prolongs the service life of the wells, which is very valuable for improving production efficiency and economic of the oilfield.

In the Da13 block, with a reservoir depth of 4200m, average shut-in pressure is 36.5MPa, the target layer closure pressure is 79MPa, and the bottomhole flowing pressure is 22 ~37MPa. The proppant is required to withstand pressures is 42 ~57MPa. To meet the long-term conductivity of fractures, a ceramic proppant with a pressure of 69MPa is employed. The proppant design adopts staged plugs, with 40/70 mesh quartz sand (5m³ per stage) for polishing to reduce resistance. The main proppant is 30/50 mesh ceramic proppant to ensure high conductivity of the fractures.

Proppant concentration recommendation. To maximize the conductivity of fractures, it is recommended that an average proppant concentration is 7.32kg/m³. Taking a fracture with an average length of 80-100m and a height of 25-30m, the calculated sanding intensity is about 1.3m³/m (table 2). Combining the well production performance verse the sand addition scales, the recommended sanding intensity for single fracture is 1.3m³/m. The final sand addition volume of individual well fracturing design will be determined according to well logging and mud logging data and the development status of the oil layer.

Conclusion

For 20/40-, 40/70-, and 70/140-mesh quartz sand, conventional stepwise fracturing forms distinct proppant dunes near the wellbore with sparse distal coverage and low concentration. Constant-concentration fracturing mitigates this issue, achieving uniform distribution along the entire fracture length from near-well to toe regions with higher proppant coverage ratio. Compared to stepwise methods, constant-concentration

fracturing optimizes placement profiles, enhances toe-end proppant concentration, and effectively supports fracture length - improving hydrocarbon migration efficiency. Using low-viscosity slickwater throughout constant-concentration operations enables effective transport of small-mesh proppants ($\leq 40/70$ -mesh) at high injection rates, ensuring target transport distances. Unconventional reservoir stimulation is recommended to adopt: small-mesh proppants + low-viscosity slickwater + high-rate injection with moderate-to-high constant concentrations to enhance operational efficiency and production.

When the fracturing stage length is less than 110m, the increase in EUR is not significant, and fracturing costs increase substantially. When the fracturing stage length exceeds 110m, EUR significantly decreases. Balance fracturing costs and single well EUR, it is recommended to set the fracturing stage length in the middle and shallow DaHW well at 110m. To meet the long-term conductivity of fractures, a ceramic proppant with a pressure of 69MPa is employed. The proppant design adopts staged plugs, with 40/70 mesh quartz sand (5m³ per stage) for polishing to reduce resistance. The main proppant is 30/50 mesh ceramic proppant to ensure high conductivity of the fractures. Combining the well production performance verse the sand addition scales, the recommended sanding intensity for single fracture is 1.3m³/m. Compared the result of production simulations with different numbers of fractures, indicate an obviously advantage in production when the number of fractures is 75 over the same period.

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